Class 12 Mathematics

Chapter 9: Differential Equations

Practice Set A: step-wise solutions

20 questions **51** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** CBSE 2023 — 1 mark

The order and the degree of the differential equation $\left(1+3\frac{dy}{dx}\right)^2 = 4\frac{d^3y}{dx^3}$ respectively are:

(A) $1,\ \frac{2}{3}$ **(B)** $3,\ 1$ **(C)** $3,\ 3$ **(D)** $1,\ 2$

Formula / idea: Order = order of the highest derivative; degree = its highest power, once the equation is a polynomial in derivatives.

1 Find the highest order derivative.

$\frac{d^3y}{dx^3} \Rightarrow \text{Order} = 3$

2 The equation is a polynomial in derivatives; the power of $\frac{d^3y}{dx^3}$ is 1. ✓ 1

$\text{Degree} = 1$

3 Hence option (B).

Answer **(B)** $3,\ 1$

Watch out: Taking degree 2 from the square on $\left(1+3\frac{dy}{dx}\right)$; degree is read only from the highest order derivative.

2 **MCQ** CBSE 2023 — 1 mark

The degree of the differential equation $\sin x + \cos\left(\frac{dy}{dx}\right) = y^2$ is:

(A) 2 **(B)** 1 **(C)** **not defined** **(D)** 0

Formula / idea: Degree is defined only when the equation is a polynomial in its derivatives.

1 The only derivative is $\frac{dy}{dx}$, so the order is 1.

2 The term $\cos\left(\frac{dy}{dx}\right)$ is not a polynomial in $\frac{dy}{dx}$. ✓ 1

3 So the degree is not defined. Hence option (C).

Answer **(C) not defined**

Watch out: Blaming $\sin x$ or y^2; they are fine. Only a non-polynomial function of a derivative, here $\cos\left(\frac{dy}{dx}\right)$, makes the degree undefined.

3 MCQ CBSE 2023 1 mark

The integrating factor of the differential equation $(1-y^2)\frac{dx}{dy} + yx = ay, (-1 < y < 1)$ is:

(A) $\frac{1}{y^2-1}$ **(B)** $\frac{1}{\sqrt{y^2-1}}$ **(C)** $\frac{1}{1-y^2}$ **(D)** $\frac{1}{\sqrt{1-y^2}}$

Formula / idea: For $\frac{dx}{dy} + Px = Q$ (P, Q functions of y), $\text{I.F.} = e^{\int P\,dy}$.

1 Divide by $1-y^2$ to get the standard form.
$$\frac{dx}{dy} + \frac{y}{1-y^2}x = \frac{ay}{1-y^2}$$

2 Compare with $\frac{dx}{dy} + Px = Q$.
$$P = \frac{y}{1-y^2}$$

3 Integrate P.
$$\int \frac{y}{1-y^2}\,dy = -\frac{1}{2}\log(1-y^2)$$
Put $1-y^2 = t$, so $-2y\,dy = dt$.

4 Find the integrating factor. ✓ 1
$$\text{I.F.} = e^{-\frac{1}{2}\log(1-y^2)} = (1-y^2)^{-\frac{1}{2}} = \frac{1}{\sqrt{1-y^2}}$$

5 Hence option (D).

Answer **(D)** $\frac{1}{\sqrt{1-y^2}}$

Watch out: Dropping the $\frac{1}{2}$ from the integral and choosing (C).

4 MCQ NCERT Ex 9.2 Q12 1 mark

The number of arbitrary constants in the particular solution of a differential equation of third order are:

(A) 3 **(B)** 2 **(C)** 1 **(D)** 0

Formula / idea: A particular solution is got from the general solution by giving fixed values to all its arbitrary constants.

1	The general solution of a third order equation has 3 arbitrary constants.	
2	In a particular solution, every constant gets a fixed value. So none is left.	✓ 1
3	Hence option (D).	

Answer **(D)** 0

Watch out: Answering 3; that is the count for the general solution, not the particular solution.

5 MCQ CBSE SQP 2024-25 1 mark

What is the general solution of the differential equation $e^{y'} = x$?

(A) $y = x\log x + c$ **(B)** $y = x\log x - x + c$

(C) $y = x\log x + x + c$ **(D)** $y = x + c$

Formula / idea: $\int \log x\, dx = x\log x - x + c$ (by parts).

1	Take log on both sides. $y' = \log x \Rightarrow \frac{dy}{dx} = \log x$	
2	Separate and integrate. $y = \int \log x\, dx$	
3	Integrate by parts, taking 1 as the second function. $y = x\log x - \int x \cdot \frac{1}{x}\, dx = x\log x - x + c$	✓ 1
4	Hence option (B).	

Answer **(B)** $y = x\log x - x + c$

Watch out: Forgetting the $-x$ from integration by parts and choosing (A).

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): $y = ae^{3x} + be^{-2x}$ is a solution of the differential equation $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$. **Reason (R):** The number of arbitrary constants in the general solution of a differential equation is equal to its order.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) **Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

1 Differentiate y twice.
$\frac{dy}{dx} = 3ae^{3x} - 2be^{-2x}, \qquad \frac{d^2y}{dx^2} = 9ae^{3x} + 4be^{-2x}$

2 Substitute in the LHS: it is 0 for every x. So A is true.
$(9 - 3 - 6)\,ae^{3x} + (4 + 2 - 6)\,be^{-2x} = 0$

3 R is a standard fact about general solutions. So R is true.

4 A holds because y satisfies the equation on substitution. R does not explain this. ✓ 1

5 Hence option (B).

Answer **(B) Both A and R are true, but R is not the correct explanation of A.**

Watch out: Choosing (A) because both statements mention a second order equation; R says nothing about why y satisfies it.

7 **Short answer** PYQ-style 2 marks

Find the general solution of the differential equation $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$.

Formula / idea: Variable separable: bring all y terms to dy and all x terms to dx, then integrate.

1 Take out the common factor e^{-y}.
$\frac{dy}{dx} = e^{-y}\left(e^x + x^2\right)$

2 Separate the variables. ✓ ½
$e^y\,dy = \left(e^x + x^2\right)dx$

3 Integrate both sides. ✓ 1
$\int e^y\,dy = \int \left(e^x + x^2\right)dx \;\Rightarrow\; e^y = e^x + \frac{x^3}{3} + C$

4	Hence the general solution is $e^y = e^x + \dfrac{x^3}{3} + C$.	✓ ½

Answer $e^y = e^x + \dfrac{x^3}{3} + C$

Watch out: Writing $e^{x-y} = e^x - e^y$; the correct split is $e^{x-y} = e^x \cdot e^{-y}$.

Level 2

18 marks

8 **Short answer** NCERT Ex 9.3 Q20 2 marks

In a bank, principal increases continuously at the rate of $r\%$ per year. Find the value of r if ₹100 double itself in 10 years $(\log_e 2 = 0.6931)$.

Formula / idea: Rate proportional to amount: $\dfrac{dP}{dt} = kP$ gives $P = P_0 e^{kt}$.

1	Let P be the principal at time t years. Write the rate of growth. $\dfrac{dP}{dt} = \dfrac{r}{100} P$	✓ ½
2	Separate and integrate. $\dfrac{dP}{P} = \dfrac{r}{100}\,dt \Rightarrow \log P = \dfrac{rt}{100} + C$	
3	At $t = 0$, $P = 100$, so $C = \log 100$. $\log \dfrac{P}{100} = \dfrac{rt}{100}$	✓ ½
4	At $t = 10$, $P = 200$. $\log 2 = \dfrac{10r}{100} = \dfrac{r}{10}$	✓ ½
5	Solve for r. $r = 10 \log 2 = 10 \times 0.6931 = 6.931$	✓ ½
6	Hence the rate is 6.931% per year.	

Answer $r = 6.931$, **i.e. the rate is** 6.931% **per year**

Watch out: Writing $\dfrac{dP}{dt} = rP$; the rate is $r\%$, so the constant is $\dfrac{r}{100}$.

9 **Short answer** NCERT Misc. Ex. Q7 3 marks

Find the particular solution of the differential equation $(1+e^{2x})\,dy+(1+y^2)\,e^x\,dx=0$, given that $y=1$ when $x=0$.

Formula / idea: $\displaystyle\int \frac{dt}{1+t^2}=\tan^{-1}t+C$

1	Separate the variables. $\displaystyle\frac{dy}{1+y^2}=-\frac{e^x}{1+e^{2x}}\,dx$	✓ ½
2	In the RHS put $e^x=t$, so $e^x\,dx=dt$. $\displaystyle\int\frac{dy}{1+y^2}=-\int\frac{dt}{1+t^2}$	✓ ½
3	Integrate both sides and put back $t=e^x$. $\tan^{-1}y=-\tan^{-1}(e^x)+C\ \Rightarrow\ \tan^{-1}y+\tan^{-1}(e^x)=C$	✓ 1
4	Put $x=0, y=1$. $\tan^{-1}1+\tan^{-1}1=C\ \Rightarrow\ C=\frac{\pi}{4}+\frac{\pi}{4}=\frac{\pi}{2}$	✓ ½
5	Hence the particular solution is $\tan^{-1}y+\tan^{-1}(e^x)=\frac{\pi}{2}$, i.e. $y=e^{-x}$. $\frac{\pi}{2}-\tan^{-1}e^x=\cot^{-1}e^x=\tan^{-1}e^{-x}$	✓ ½

Answer $\tan^{-1}y+\tan^{-1}(e^x)=\dfrac{\pi}{2}$, **i.e.** $y=e^{-x}$

Watch out: Putting $e^{2x}=t$ instead of $e^x=t$; then $e^x\,dx$ in the numerator does not become dt.

10 **Short answer** PYQ-style 3 marks

Find the general solution of the differential equation $x\dfrac{dy}{dx}+y=x\cos x+\sin x,\ (x\neq 0)$.

Formula / idea: For $\dfrac{dy}{dx}+Py=Q$: $y\cdot\text{I.F.}=\displaystyle\int Q\cdot\text{I.F.}\,dx+C$, where $\text{I.F.}=e^{\int P\,dx}$.

1	Divide by x to get the standard form. $\displaystyle\frac{dy}{dx}+\frac{1}{x}y=\cos x+\frac{\sin x}{x}$	✓ ½
2	Here $P=\dfrac{1}{x}$. Find the integrating factor. $\text{I.F.}=e^{\int\frac{1}{x}dx}=e^{\log x}=x$	✓ ½
3	Write the solution formula. $y\cdot x=\displaystyle\int x\left(\cos x+\frac{\sin x}{x}\right)dx=\int(x\cos x+\sin x)\,dx$	✓ ½

4 Integrate $x\cos x$ by parts. ✓ ½

$$\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x$$

5 Add $\int \sin x\,dx = -\cos x$. ✓ ½

$$xy = x\sin x + \cos x - \cos x + C = x\sin x + C$$

6 Hence the general solution is $y = \sin x + \dfrac{C}{x}$. ✓ ½

Answer $xy = x\sin x + C$, **i.e.** $y = \sin x + \dfrac{C}{x}$

Watch out: Taking $P = 1$ and $Q = x\cos x + \sin x$ without first dividing the whole equation by x.

11 **Short answer** CBSE 2023 3 marks

Solve the differential equation $(x^2 - y^2)\,dx + 2xy\,dy = 0$.

Formula / idea: Homogeneous equation $\dfrac{dy}{dx} = g\left(\dfrac{y}{x}\right)$: put $y = vx$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$.

1 Write $\dfrac{dy}{dx}$. Numerator and denominator are both of degree 2, so it is homogeneous. ✓ ½

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

2 Put $y = vx$. ✓ ½

$$v + x\frac{dv}{dx} = \frac{v^2x^2 - x^2}{2vx^2} = \frac{v^2 - 1}{2v}$$

3 Move v to the right. ✓ ½

$$x\frac{dv}{dx} = \frac{v^2 - 1 - 2v^2}{2v} = -\frac{1 + v^2}{2v}$$

4 Separate the variables. ✓ ½

$$\frac{2v}{1 + v^2}\,dv = -\frac{dx}{x}$$

5 Integrate both sides. ✓ ½

$$\log(1 + v^2) = -\log|x| + \log|C| \;\Rightarrow\; x(1 + v^2) = C$$

6 Put $v = \dfrac{y}{x}$. Hence the general solution is $x^2 + y^2 = Cx$. ✓ ½

$$x\left(1 + \frac{y^2}{x^2}\right) = C \;\Rightarrow\; x^2 + y^2 = Cx$$

Answer $x^2 + y^2 = Cx$

Watch out: Getting the sign of $x^2 - y^2$ wrong when moving it across: $\dfrac{dy}{dx} = \dfrac{y^2 - x^2}{2xy}$, not $\dfrac{x^2 - y^2}{2xy}$.

12 **Short answer** PYQ-style 2 marks

Verify that $y = A\cos(\log x) + B\sin(\log x)$ is a solution of the differential equation $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

Formula / idea: A function is a solution if it satisfies the equation identically.

1 Differentiate w.r.t. x (chain rule). ✓ ½
$$\frac{dy}{dx} = \frac{-A\sin(\log x) + B\cos(\log x)}{x}$$

2 Multiply by x. ✓ ½
$$x\frac{dy}{dx} = -A\sin(\log x) + B\cos(\log x)$$

3 Differentiate again w.r.t. x (product rule on the left). ✓ ½
$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} = \frac{-A\cos(\log x) - B\sin(\log x)}{x} = -\frac{y}{x}$$

4 Multiply by x and rearrange. Hence verified. ✓ ½
$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$$

Answer $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$ **holds, so the function is a solution.**

Watch out: Forgetting the factor $\dfrac{1}{x}$ from the chain rule when differentiating $\cos(\log x)$.

13 **Short answer** Board-pattern 2 marks

Find the general solution of the differential equation $\dfrac{dy}{dx} = x\log x$, $(x > 0)$.

Formula / idea: Integration by parts: $\displaystyle\int u\,v\,dx = u\int v\,dx - \int\left(u'\int v\,dx\right)dx$.

1 Separate and integrate. ✓ ½
$$dy = x\log x\,dx \;\Rightarrow\; y = \int x\log x\,dx$$

2 Integrate by parts, taking $\log x$ as the first function. ✓ ½
$$y = \log x\cdot\frac{x^2}{2} - \int\frac{1}{x}\cdot\frac{x^2}{2}\,dx$$

3 Simplify and integrate. ✓ ½
$$y = \frac{x^2}{2}\log x - \int\frac{x}{2}\,dx = \frac{x^2}{2}\log x - \frac{x^2}{4} + C$$

4	Hence the general solution is $y = \dfrac{x^2}{2}\log x - \dfrac{x^2}{4} + C$.	✓ ½

Answer $y = \dfrac{x^2}{2}\log x - \dfrac{x^2}{4} + C$

Watch out: Taking x as the first function; then $\log x$ must be integrated, which makes it harder.

14 **Short answer** Board-pattern 3 marks

Find the particular solution of the differential equation $\dfrac{dy}{dx} + y = \cos x - \sin x$, given that $y = 3$ when $x = 0$.

Formula / idea: $\displaystyle\int e^x\,[f(x) + f'(x)]\,dx = e^x f(x) + C$

1	Compare with $\dfrac{dy}{dx} + Py = Q$. $P = 1, \quad Q = \cos x - \sin x$	✓ ½
2	Find the integrating factor. $\text{I.F.} = e^{\int 1\,dx} = e^x$	✓ ½
3	Write the solution formula. $y\,e^x = \displaystyle\int e^x\,(\cos x - \sin x)\,dx + C$	✓ ½
4	Take $f(x) = \cos x$, so $f'(x) = -\sin x$. $y\,e^x = e^x \cos x + C$	✓ ½
5	Put $x = 0, y = 3$. $3 = 1 + C \;\Rightarrow\; C = 2$	✓ ½
6	Divide by e^x. Hence the particular solution is $y = \cos x + 2e^{-x}$. $y = \cos x + 2e^{-x}$	✓ ½

Answer $y = \cos x + 2e^{-x}$

Watch out: Forgetting to divide C by e^x too: the answer is $\cos x + 2e^{-x}$, not $\cos x + 2$.

Level 3

25 marks

15 **Long answer** NCERT Ex 9.4 Q1 5 marks

Show that the differential equation $(x^2 + xy)\,dy = (x^2 + y^2)\,dx$ is homogeneous and solve it.

Formula / idea: Homogeneous equation: $F(\lambda x, \lambda y) = \lambda^0 F(x, y)$; then put $y = vx$.

1	Write $\dfrac{dy}{dx} = F(x, y)$. $\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{x^2 + xy} = F(x, y)$	✓ ½
2	Replace x, y by $\lambda x, \lambda y$. So F is homogeneous of degree 0. $F(\lambda x, \lambda y) = \dfrac{\lambda^2(x^2 + y^2)}{\lambda^2(x^2 + xy)} = \lambda^0 F(x, y)$	✓ ½
3	Put $y = vx$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. $v + x\dfrac{dv}{dx} = \dfrac{1 + v^2}{1 + v}$	✓ ½
4	Move v to the right. $x\dfrac{dv}{dx} = \dfrac{1 + v^2 - v - v^2}{1 + v} = \dfrac{1 - v}{1 + v}$	✓ ½
5	Separate the variables. $\dfrac{1 + v}{1 - v}\,dv = \dfrac{dx}{x}$	✓ ½
6	Split the fraction. $\dfrac{1 + v}{1 - v} = -1 + \dfrac{2}{1 - v}$ $1 + v = 2 - (1 - v)$	✓ ½
7	Integrate both sides. $-v - 2\log\lvert 1 - v\rvert = \log\lvert x\rvert + C_1$	✓ 1
8	Put $v = \dfrac{y}{x}$. $-\dfrac{y}{x} - 2\log\left\lvert\dfrac{x - y}{x}\right\rvert = \log\lvert x\rvert + C_1$	✓ ½
9	Open the log and collect the log terms on the left. $\log\lvert x\rvert - 2\log\lvert x - y\rvert = \dfrac{y}{x} + C_1 \;\Rightarrow\; \log\dfrac{\lvert x\rvert}{(x - y)^2} = \dfrac{y}{x} + C_1$	

10 Take exponentials. Hence the general solution is $(x-y)^2 = Cx\,e^{-y/x}$. ✓ ½

$(x-y)^2 = Cx\,e^{-y/x}$

$C = \pm e^{-C_1}$ is an arbitrary constant.

Answer $(x-y)^2 = Cx\,e^{-y/x}$

Watch out: Writing $\int \frac{2}{1-v}\,dv = 2\log|1-v|$; it is $-2\log|1-v|$, because the derivative of $1-v$ is -1.

16 **Short answer** CBSE SQP 2026-27 3 marks

Find the particular solution of the differential equation $x\,dy + (y - x^2e^x)\,dx = 0$, given that $y(1) = 0$.

Formula / idea: For $\frac{dy}{dx} + Py = Q$: $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$.

1 Divide by $x\,dx$ to get the standard form. ✓ ½

$\frac{dy}{dx} + \frac{1}{x}y = xe^x$

2 Here $P = \frac{1}{x}$. Find the integrating factor. ✓ ½

$\text{I.F.} = e^{\int \frac{1}{x}dx} = e^{\log x} = x$

3 Write the solution formula. ✓ ½

$y \cdot x = \int x \cdot xe^x\,dx = \int x^2e^x\,dx$

4 Integrate by parts twice. ✓ ½

$xy = x^2e^x - 2\int xe^x\,dx$

$= x^2e^x - 2(xe^x - e^x) + C = e^x(x^2 - 2x + 2) + C$

5 Put $x = 1, y = 0$. ✓ ½

$0 = e(1 - 2 + 2) + C \;\Rightarrow\; C = -e$

6 Hence the particular solution is $xy = e^x(x^2 - 2x + 2) - e$. ✓ ½

$xy = e^x(x^2 - 2x + 2) - e$

Answer $xy = e^x(x^2 - 2x + 2) - e$

Watch out: Taking $Q = x^2e^x$; after dividing by x the right side is xe^x.

17 **Long answer** CBSE SQP 2025-26 5 marks

Find the particular solution of the differential equation $2y\,e^{x/y}\,dx + \left(y - 2x\,e^{x/y}\right)dy = 0$, given that $y(0) = 1$.

Formula / idea: When $\dfrac{dx}{dy} = F\left(\dfrac{x}{y}\right)$, put $x = vy$, $\dfrac{dx}{dy} = v + y\dfrac{dv}{dy}$.

1	Write $\dfrac{dx}{dy} = F(x, y)$. $\dfrac{dx}{dy} = \dfrac{2x\,e^{x/y} - y}{2y\,e^{x/y}} = F(x, y)$	✓ ½
2	Replace x, y by $\lambda x, \lambda y$; $\dfrac{x}{y}$ does not change. So F is homogeneous of degree 0. $F(\lambda x, \lambda y) = \lambda^0 F(x, y)$	✓ ½
3	Put $x = vy$, $\dfrac{dx}{dy} = v + y\dfrac{dv}{dy}$. $v + y\dfrac{dv}{dy} = \dfrac{2vy\,e^v - y}{2y\,e^v} = \dfrac{2v\,e^v - 1}{2e^v}$	✓ ½
4	Move v to the right. $y\dfrac{dv}{dy} = \dfrac{2v\,e^v - 1}{2e^v} - v = -\dfrac{1}{2e^v}$	✓ ½
5	Separate the variables. $2e^v\,dv = -\dfrac{dy}{y}$	✓ ½
6	Integrate both sides. $2e^v = -\log\lvert y\rvert + C$	✓ 1
7	Put $v = \dfrac{x}{y}$. $2e^{x/y} + \log\lvert y\rvert = C$	✓ ½
8	Put $x = 0, y = 1$. $2e^0 + \log 1 = C \;\Rightarrow\; C = 2$	✓ ½
9	Hence the particular solution is $2e^{x/y} + \log\lvert y\rvert = 2$.	✓ ½

Answer $2e^{x/y} + \log\lvert y\rvert = 2$

Watch out: Putting $y = vx$ out of habit; here $\dfrac{x}{y}$ appears, so $x = vy$ is the easy substitution.

18 **Short answer** CBSE 2023 — 3 marks

Find the general solution of the differential equation $(xy - x^2)\,dy = y^2\,dx$.

Formula / idea: Homogeneous equation: put $y = vx$, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$.

1. Write $\dfrac{dy}{dx}$. Numerator and denominator are both of degree 2, so it is homogeneous. ✓ ½
$$\frac{dy}{dx} = \frac{y^2}{xy - x^2}$$

2. Put $y = vx$. ✓ ½
$$v + x\frac{dv}{dx} = \frac{v^2x^2}{vx^2 - x^2} = \frac{v^2}{v - 1}$$

3. Move v to the right. ✓ ½
$$x\frac{dv}{dx} = \frac{v^2 - v^2 + v}{v - 1} = \frac{v}{v - 1}$$

4. Separate the variables. ✓ ½
$$\left(1 - \frac{1}{v}\right)dv = \frac{dx}{x}$$

5. Integrate both sides. ✓ ½
$$v - \log|v| = \log|x| + C$$

6. Put $v = \dfrac{y}{x}$; $\log|x|$ cancels. Hence $\dfrac{y}{x} - \log|y| = C$. ✓ ½
$$\frac{y}{x} - \log|y| + \log|x| = \log|x| + C \;\Rightarrow\; \frac{y}{x} - \log|y| = C$$

Answer $\dfrac{y}{x} - \log|y| = C$

Watch out: Writing $\log\left|\dfrac{y}{x}\right| = \dfrac{\log|y|}{\log|x|}$; the correct rule is $\log|y| - \log|x|$.

19 **Long answer** PYQ-style — 5 marks

Find the particular solution of the differential equation $y\,e^y\,dx = \left(y^3 + 2x\,e^y\right)dy$, given that $x = 0$ when $y = 1$.

Formula / idea: For $\dfrac{dx}{dy} + Px = Q$: $x \cdot \text{I.F.} = \displaystyle\int Q \cdot \text{I.F.}\,dy + C$, where $\text{I.F.} = e^{\int P\,dy}$.

1. Write $\dfrac{dx}{dy}$ and split the fraction. ✓ ½
$$\frac{dx}{dy} = \frac{y^3 + 2x\,e^y}{y\,e^y} = y^2e^{-y} + \frac{2x}{y}$$

2. Rearrange: it is linear in x. ✓ ½
$$\frac{dx}{dy} - \frac{2}{y}\,x = y^2e^{-y}$$

3 Compare with $\frac{dx}{dy} + Px = Q$.

$P = -\frac{2}{y}, \quad Q = y^2 e^{-y}$

4 Find the integrating factor. ✓ 1

$\text{I.F.} = e^{\int -\frac{2}{y} dy} = e^{-2\log y} = \frac{1}{y^2}$

5 Write the solution formula. ✓ 1

$x \cdot \frac{1}{y^2} = \int y^2 e^{-y} \cdot \frac{1}{y^2}\, dy + C$

6 Integrate. ✓ ½

$\frac{x}{y^2} = \int e^{-y}\, dy = -e^{-y} + C$

7 Put $x = 0, y = 1$. ✓ ½

$0 = -e^{-1} + C \;\Rightarrow\; C = \frac{1}{e}$

8 Substitute C. ✓ ½

$\frac{x}{y^2} = \frac{1}{e} - e^{-y}$

9 Hence the particular solution is $x = y^2\left(e^{-1} - e^{-y}\right)$. ✓ ½

$x = y^2\left(e^{-1} - e^{-y}\right)$

Answer $x = y^2\left(e^{-1} - e^{-y}\right)$

Watch out: Trying to write it as $\frac{dy}{dx} + Py = Q$; it is linear only in x, so use $\frac{dx}{dy}$.

20 Case study Board-pattern — 4 marks

Case Study: The population of a village increases continuously at a rate proportional to the number of its inhabitants present at any time. The population was 20,000 in the year 2019 and 25,000 in the year 2024. Measure the time t in years from 2019.

(i) Write the differential equation for the population P at time t and state its order and degree. — 1 mark

(ii) Solve it to express P in terms of t and find the constant k. — 1 mark

(iii) In which year will the population be 31,250? — 2 marks

Formula / idea: Rate proportional to amount: $\frac{dP}{dt} = kP$ gives $P = P_0 e^{kt}$.

(i) — 1 mark

1 Rate of increase is proportional to P, with constant $k > 0$. ✓ ½

$\frac{dP}{dt} = kP$

2 Highest derivative is $\frac{dP}{dt}$, with power 1.
$\text{Order} = 1, \quad \text{Degree} = 1$ ✓ ½

Answer (i) $\frac{dP}{dt} = kP$; **order 1, degree 1**

(ii) 1 mark

1 Separate and integrate.
$\frac{dP}{P} = k\,dt \;\Rightarrow\; \log P = kt + C$

2 At $t = 0$, $P = 20000$, so $C = \log 20000$. ✓ ½
$\log \frac{P}{20000} = kt \;\Rightarrow\; P = 20000\,e^{kt}$

3 At $t = 5$ (year 2024), $P = 25000$.
$25000 = 20000\,e^{5k} \;\Rightarrow\; e^{5k} = \frac{5}{4}$

4 Take log on both sides. ✓ ½
$k = \frac{1}{5}\log\frac{5}{4}$

Answer (ii) $P = 20000\,e^{kt}$, $k = \frac{1}{5}\log\frac{5}{4}\ (\approx 0.0446)$

(iii) 2 marks

1 Put $P = 31250$. ✓ ½
$31250 = 20000\,e^{kt} \;\Rightarrow\; e^{kt} = \frac{25}{16}$

2 Write $\frac{25}{16}$ using $e^{5k} = \frac{5}{4}$. ✓ ½
$\frac{25}{16} = \left(\frac{5}{4}\right)^2 = \left(e^{5k}\right)^2 = e^{10k}$

3 Compare powers. ✓ ½
$e^{kt} = e^{10k} \;\Rightarrow\; t = 10$

4 Count 10 years from 2019. Hence the population will be 31,250 in the year 2029. ✓ ½
$2019 + 10 = 2029$

Answer (iii) $t = 10$ **years, i.e. in the year 2029**

Watch out: Taking $t = 2024$ instead of $t = 5$; time must be counted from the starting year 2019.